

I want to follow the evolution in time of swell for a specific frequency range. In order to do that I need to calculate the “Equivalent Wave Height=EHW” from the wave spectra, like this:

$$EHW_{f_1-f_2} = 4 \sqrt{\int_0^{2\pi} \int_{f_1}^{f_2} SPEC(\omega, \theta) d\omega d\theta}$$

Where SPEC is the 2-dimensional wave spectrum, ω is the angular frequency and θ is the wave direction.

Let's choose the frequency range to be $f_1 = 0.04595011$ and $f_2 = 0.05559963$ (17.9858 - 21.762sec)

Take the wave spectra from:

https://thredds.met.no/thredds/catalog/windsurfer/mywavewam3km_spectra/1995/02/catalog.html
https://thredds.met.no/thredds/catalog/windsurfer/mywavewam3km_spectra/1995/02/catalog.html?dataset=windsurfer/mywavewam3km_spectra/1995/02/SPC1995020100.nc

And then plot the contours of $EHW_{f_1-f_2}$ for the day 19950201